

Roll No.

DETF-01

**B. TECH. (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

SOLAR ENERGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What is solar collector and why should one go for it ? What are the advantages of fiat plate collectors ? Explain the working principle of fiat plate collector with a neat sketch. (CO2)

OR

- (b) Why should one go for concentrator collector ? Explain the working principle of compound parabolic concentrator with a neat sketch. Also write its industrial application. (CO2)

2. (a) Write energy balance equations for fiat plate collector and hence derive the expression for temperature of fluid at the outlet of fiat plate collector. (CO2)

OR

- (b) Write energy balance equations for fully covered photovoltaic thermal compound parabolic concentrator collector and hence derive the expression for temperature of fluid at the outlet of fully covered photovoltaic thermal compound parabolic concentrator collector. (CO2)

P. T. O.

(2)

3. (a) Write energy balance equations for evacuated tubular collector and hence derive the expression for temperature of fluid at the outlet of evacuated tubular collector. (CO2)

OR

- (b) Explain the solar radiation measuring instruments with a neat sketch. (CO1)

4. (a) Explain various Sun-earth angles with neat sketch. (CO1)

OR

- (b) Explain the electrical analogy of thermal system with a neat sketch. Also, explain the estimation of overall heat transfer coefficient with a suitable example. (CO1)

5. (a) Explain conduction, convection, radiation and evaporation. (CO1)

OR

- (b) Explain greenhouse effect in detail with suitable example. Draw thermal circuit for any solar system and hence find overall heat transfer coefficient for the solar system. (CO1)